The Digits of 6174 at Every Length: Two Liftings of a Universal Kaprekar Fixed Point

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Abstract

The Kaprekar routine sends every four-digit number to 6174. This paper asks whether the digits of 6174 keep that role at every length: for each d, is there an arrangement of those digits — extended with zeros or by repetition — that some single rule, uniform in d, makes a universal fixed point, drawn to by every admissible input? The maps allowed are the generalized Kaprekar routines, which sort the digits and subtract one rearrangement from another, a family of $d!\,(d!-1)$ maps at each length. A finite-state criterion reduces universality to fixed-point uniqueness together with acyclicity, and an exhaustive census at $d \leq 6$ — 0, 4, 33, and 506 universal full-variable fixed points — singles out, among the thirty-three at $d = 5$, the integer 60714, the digits of 6174 with a zero, as the unique one surviving to $d = 6$. In the zero direction the question is then settled: an explicit coefficient-preserving ladder makes 60714, padded with zeros, a universal fixed point at $d = 5, 6$ and a universal one off an explicit escape class that collapses to 0 — the analogue of the multiples of 1111 the classical map itself sends to 0 — at every larger length, all proven through an absorbing set the padding creates. In the repetition direction — the multisets $\{1, 4, 6, 7\}^m$ — a folding operation preserves the fixed-point equation at every multiplicity and, by complete enumeration, preserves universality often: all eight universal rules at $d = 4$ double to $d = 8$, 144 of 312 at $d = 8$ double to $d = 16$, and the folds of the rule fixing 1746 are universal at multiplicities 2, 4, 5, the last over all 10,014,995 multisets at $d = 20$, where only proxy evidence existed before. Whether universality holds at every m is open, but the surviving folds share an absorbing structure of their own — a sign-coherent core, reached in a few steps, on which the dynamics funnels to the fixed point through a finite quotient — reducing the sharpest case to two uniformities in m. The two directions differ for one reason, which we prove: zero-padding has an absorbing set to drive orbits home, while the base case of the repetition direction, Kaprekar's own theorem, carries no polynomial Lyapunov certificate below degree seven.

1 Introduction

Kaprekar observed in 1949 that the map which sorts the digits of a four-digit integer into descending and ascending order and subtracts sends every input with at least two distinct digits to 6174, in at most seven steps [Kaprekar 1955]. The three-digit analogue converges to 495 [Trigg 1974]; the five-digit analogue converges to nothing, every orbit falling into one of three cycles [Prichett 1981]. For seventy-five years these facts have been verified the same way they were discovered, by finite enumeration. They belong to a family far larger than the classical map, and within that family the sharper and more surprising statements are constructive ones.

The classical routine is one member of a large family. Fix a length d, pad integers to d digits, and let any ordered pair of distinct permutations rearrange the sorted digit string before the subtraction;

this yields $d!\,(d!-1)$ maps at each length, of which Kaprekar's is one. The question this paper sets out to answer is whether the digits of 6174 keep their role across this family at every length. Precisely: *for each d, is there an arrangement of those digits — extended either with zeros or by repetition — and a single rule, uniform in d, for which that arrangement is a universal fixed point, reached by every admissible input?* The question has two directions, set by the two ways of extending the digits, and they behave very differently.

The generalized family has a growing literature, most of it fixing the length or the base and classifying the dynamics that result. Nuez studies parametric and algebraic families of Kaprekar-type maps and their symmetries [Nuez 2021, Nuez 2021b, Nuez 2022]; Kay and Downes-Ward determine the fixed points and cycles of the base-b transformation in odd and even bases [Kay 2022, Kay 2024]; Prichett's determination of the decadic constants, with its five-digit precursor, settles the classical fixed lengths [Prichett 1981]; Devlin and Zeng bound the four-digit reaching time [Devlin 2021]; Peterson and Pulapaka survey the routine across bases [Peterson 2008]; and Dahl analyzes its basins and attractor entropy [Dahl 2026]. This paper takes the orthogonal cut: it fixes the *fixed point* — the digits of 6174 — and varies the length, asking for a single rule, uniform across lengths, that keeps that fixed point universal. The object of study is the cross-dimensional lifting, not the classification at any one length.

Take the zeros first. Universality — every admissible input reaching the fixed point — is rare, and universality that survives a change of length is rarer. An exhaustive census at $d \le 6$ (Section 3) finds four universal full-variable fixed points at $d = 4$, thirty-three at $d = 5$, five hundred six at $d = 6$; of the thirty-three, exactly one — 60714, the digits of 6174 with a zero — is universal again at $d = 6$. That survivor extends to every length (Section 4): a rule given explicitly at each $d \ge 5$, the rungs linked by a coefficient-preserving ladder, under which 60714 padded with zeros is universal. The proof runs on an absorbing set the padding zeros create; it settles the zero direction — the escape class that surfaces for $d \ge 7$, where some inputs collapse to 0, is the exact analogue of the classical map's own collapsing inputs and is fully characterized, not a gap — and it is worth isolating because it is the mechanism the other direction lacks.

Now the repetition. The same digits can be taken with multiplicity m instead of padded — the multiset $\{1, 4, 6, 7\}^m$, a $4m$-digit problem — and universal fixed points of $\{1, 4, 6, 7\}^m$ exist at every $m \le 6$, known by enumeration and extended and sharpened here, including the first complete-basin verification at $d = 20$. Whether they exist at every m is the open conjecture. Section 5 introduces a folding operation that makes the algebraic half of the question a theorem at all m, and Section 6 reports the central computational finding of this paper: folding preserves universality often — every measured doubling step has survival fraction 100% or 46% — and the folds of one four-digit rule give fully verified universal fixed points at multiplicities $1, 2, 4, 5$, with a single twelve-digit rule covering 3 and 6. Section 6 also finds what the surviving folds have in common, and it is the structural heart of the second direction: on a sign-coherent core that every orbit reaches within a few steps, a fold collapses to a finite pair system that funnels to its fixed point — the first absorbing mechanism the repetition direction has had, and the direct analogue of the zeros that drive Section 4. The pattern of *which* folds survive is nonetheless irregular, and Section 7 measures what turning this abundance into a proof for all m would cost: the base case — Kaprekar's own theorem — carries no polynomial Lyapunov certificate below degree seven (exact Farkas certificates through degree six; an explicit degree-seven witness above). That is the price of the missing mechanism, and it is the one reason the two directions differ — zero-padding drives orbits home for free, repetition would need its base case to do the same and the base case cannot do it cheaply. Closing the gap, for the one tower where the coherent mechanism is already in hand, is the concrete open problem the paper ends on.

Throughout, claims carry one of four labels. **Proven** means a mathematical proof appears here. **Computed** means a finite exhaustive enumeration, machine-verified; unless marked otherwise, every such enumeration was re-run independently for this paper (Section 8 details the few exceptions, marked [S], which are taken from the project's computational log). **Evidence** means a claim that passes every feasible test but whose state space exceeds exhaustive reach. **Conjectured** means exactly that.

2 Rules, coefficients, and the universality criterion

Fix $d \geq 2$. For an integer n with at most d digits, write $\mathrm{sort}_\downarrow(n) = (s_0, \ldots, s_{d-1})$ for its digits padded to length d and sorted descending. For a permutation $\tau \in S_d$ let $\tau \cdot s$ denote the integer $\sum_k 10^{d-1-k} s_{\tau(k)}$.

Definition 2.1. *For an ordered pair $(\pi, \sigma) \in S_d \times S_d$ with $\pi \neq \sigma$, the rule $K_{\pi,\sigma}$ acts by*

$$K_{\pi,\sigma}(n) \;=\; \big|\, \pi \cdot \mathrm{sort}_\downarrow(n) \;-\; \sigma \cdot \mathrm{sort}_\downarrow(n) \,\big|.$$

The classical Kaprekar map is the pair (identity, reversal).

A rule is conveniently displayed by labelling the sorted positions $a = s_0,\ b = s_1,\ c = s_2, \ldots$ and writing the two rearrangements as letter-strings. The classical map at $d = 4$ is `abcd` − `dcba`: on the digits of 6174, sorted to $(a, b, c, d) = (7, 6, 4, 1)$, it computes $7641 - 1467 = 6174$. Rules quoted in the appendices use this notation.

Since the rule reads n only through its sorted digits, orbits live on digit multisets, and all state counts below are multiset counts. Expanding the difference,

$$K_{\pi,\sigma}(n) = \Big|\sum_{i=0}^{d-1} c_i\, s_i\Big|, \qquad c_i = 10^{\,d-1-\pi^{-1}(i)} - 10^{\,d-1-\sigma^{-1}(i)}, \qquad \sum_i c_i = 0.$$

The coefficient vector c determines the dynamics completely and is how rules are specified below; note that c and $-c$ induce the same map, so rules come in sign pairs with identical dynamics.

Lemma 2.2. *(Proven.) (a) $K_{\pi,\sigma}(n) \equiv 0 \pmod 9$ for every rule and every input. (b) The rules with $c_i \neq 0$ for all i — the* full-variable *rules — number $d!\, D_d$, where D_d is the number of derangements of d letters:* 12*,* 216*,* 5,280*,* 190,800 *at $d = 3, 4, 5, 6$.*

Proof. (a) Each c_i is a difference of two powers of 10, and $10^a \equiv 1 \pmod 9$. (b) $c_i = 0$ exactly when $\pi^{-1}(i) = \sigma^{-1}(i)$, so c has no zero entry exactly when $\sigma^{-1}\pi$ is fixed-point-free; for each of the $d!$ choices of π there are D_d such σ. ■

Two integer invariants grade a rule against a fixed point F with sorted digits $(f_0, \ldots, f_{d-1})$. The *rank* $\mathrm{sv}(K) = \#\{i : c_i \neq 0\}$ counts the sorted positions the rule reads; the *effective rank* $\mathrm{sv}_F(K) = \#\{i : c_i \neq 0 \text{ and } f_i \neq 0\}$ counts those that actually carry weight at F. A rule can be full-variable while $\mathrm{sv}_F < d$; the gap is invisible in the equation $K(F) = F$ and decisive for everything else (Section 3).

Definition 2.3. *F is* universal *for K at length d if $K(F) = F$ and every admissible input reaches F under iteration of K. Repdigits are always inadmissible (every rule sends them to 0). In the small-length censuses of Section 3 the ninety near-repdigits (one digit occurring $d-1$ times) are also* *excluded, following the classical convention; in the duplication chain of Sections 5–6 only repdigits are excluded. Each count below states its convention.*

Everything in the paper passes through one elementary fact.

Lemma 2.4 (universality criterion). *(Proven.) Let T be the map $X \mapsto \mathrm{sort}_{\downarrow}K(X)$ on the admissible multisets together with the absorbing state 0 — the common image of every repdigit, fixed by every rule — and let $F \neq 0$ be a fixed point of T. Then F is universal (reached by every admissible input) if and only if no fixed point other than F is reachable from an admissible input — in particular 0 is unreachable — and no admissible orbit is eventually periodic with period ≥ 2.*

Proof. The adjoined 0 makes T a genuine self-map of a finite set, so every admissible forward orbit is eventually periodic. If F is universal, then any reachable fixed point G (including 0) or any reachable period-≥ 2 cycle C would trap every admissible input flowing into it, none of which could then reach F — a contradiction; so neither exists. Conversely, each admissible orbit ends in a periodic set, which by hypothesis is neither a period-≥ 2 cycle nor any fixed point but F, hence is F. ∎

The criterion makes universality decidable by finite search and splits its failure into three modes — escape to 0, a competing fixed point, or a cycle — all of which recur below; Section 5 exhibits a rule with two fixed points and a rule with one fixed point plus cycles, and the escape mode, empty in the $d \leq 6$ censuses of Section 3, is exactly the escape class that reappears for the zero-padding chain at every $d \geq 7$ (Section 4).

3 The spectrum at small lengths

A fixed point can command anything from nothing to everything. It is useful to fix the scale before populating it.

- **L0 (collapse).** Every admissible orbit reaches 0; the rule fixes nothing worth the name.
- **L1–L2 (fixed, partial).** $K(F) = F$ but the basin is a strict fraction of the space — possibly tiny, possibly large, in the worst case erratically either.
- **L3 (degenerate universal).** F is universal but $\mathrm{sv}_F(K) < d$: the universality is that of a lower-rank map wearing d coordinates.
- **L4 (full universal).** F is universal with $\mathrm{sv}_F(K) = d$.
- **L5 (transcendent).** F, or an explicitly described family, is universal at infinitely many lengths under a fixed lifting recipe. Transcendence is *orthogonal* to the L3/L4 rank distinction, and the two liftings sit on opposite sides of it: the proven transcendent object of Section 4, 60714, is degenerate ($\mathrm{sv}_F < d$), while the conjecturally transcendent duplication family of Section 5 is full ($\mathrm{sv}_F = d$). Which side a fixed point falls on is exactly what decides whether its transcendence comes with a proof.

Proposition 3.1 (collapse is inhabited). *(Computed.) At $d = 4$, the rule with $c = (0, 0, 9, -9)$ — nine times the gap between the two smallest sorted digits — sends all 615 admissible multisets to 0. There are exactly 84 collapse rules among the 552 rules at $d = 4$, and 6 among the 30 at $d = 3$.*

At the other end of L0–L2, this paper's census found that no fixed point of any rule at $d = 4$ has a one-multiset basin (Computed): strict bareness does not occur at four digits. What occurs instead, abundantly, is the partial fixed point, and the duplication family of Section 5 will furnish the canonical example of how wildly partial basins can behave inside one algebraically uniform family.

Theorem 3.2 (census). *(Computed; exhaustive over all full-variable rules and all admissible multisets, near-repdigits excluded.) The universal full-variable fixed points at $d = 3, 4, 5, 6$ number* 0*,* 4*,* 33*,* 506*.*

1. *At $d = 3$ there are none. The universal fixed points over all thirty rules are* 45*,* 450*,* 495*, each universal for a sign-flip pair of rank-*2 *rules ($c = \pm(99, 0, -99)$ for* 495*, $\pm(9, 0, -9)$ for* 45*, $\pm(90, 0, -90)$ for* 450*); their effective ranks are $\mathrm{sv}_F = 2$,* 1*,* 1*. The classical three-digit constant is a degenerate universal.*
2. *At $d = 4$ the four are* $1746, 2538, 5382, 6174$*, each universal for exactly one sign-flip pair of rules:*

$$1746 : \pm(9, -900, 900, -9) \qquad 2538 : \pm(90, 999, -999, -90)$$

$$5382 : \pm(900, -9, 9, -900) \qquad 6174 : \pm(999, 90, -90, -999).$$

   They form two anagram clusters, $\{1, 4, 6, 7\}$ and $\{2, 3, 5, 8\}$, and all have digit sum $\equiv 0 \pmod 9$ (as Lemma 2.2(a) forces for any fixed point).
3. *At $d = 5$ the thirty-three are listed in Appendix A. Four are degenerate at their fixed point despite full-variable rules — exactly the four whose sorted form carries a zero digit, since under a full-variable rule a zero digit drops out of the effective rank:* 54 *($(5, 4, 0, 0, 0)$, $\mathrm{sv}_F = 2$),* 3753 *($\mathrm{sv}_F = 4$), and the two arrangements of $\{0, 1, 4, 6, 7\}$ —* 60417 *and* 60714*, both with sorted form $(7, 6, 4, 1, 0)$ and $\mathrm{sv}_F = 4$. The other twenty-nine have $\mathrm{sv}_F = 5$. That* 60714 *sits among the degenerate four is not a footnote: the integer this paper follows to every length is, already at its native length, rank-deficient — universal for $c = (9900, 9, 90, -9000, -999)$ (in arrangement form $71460 - 10746 = 60714$), but acting through four coordinates, not five. Its single zero is the reason, and Section 4 shows the same zero is the reason it transcends.*
4. *At $d = 6$ the five hundred six stratify by zero count as $205, 240, 53, 8$ (zero through three zeros) and by digit sum as $8, 156, 244, 96, 2$ across sums $9, 18, 27, 36, 45$.*

The count matters less than what sits inside it. The classical map is not full-variable at odd lengths — its borrow chain zeroes the middle coefficient, $c = (99, 0, -99)$ at $d = 3$ — so the classical failure at $d = 5$ is the failure of a rank-4 map asked a rank-5 question. The thirty-three show the question itself has answers.

Theorem 3.3 (cross-dimensional selection). *(Computed.) Of the thirty-three universal full-variable fixed points at $d = 5$, exactly one —* 60714 *— is again a universal full-variable fixed point at $d = 6$ under some rule. The $\{0, 0, 1, 4, 6, 7\}$ multiset carries exactly four of the* 506 *universals at $d = 6$:* 60714*,* 146070*,* 170460*,* 607140*.*

This was verified directly for this paper by intersecting the full $d = 5$ and $d = 6$ censuses. The selection is the right way to meet 60714: it is not chosen for its digits, it is the unique survivor of an exhaustive test. The next section shows the survivor goes on forever, and why.

4 Lifting by zeros: 60714 at every length

Theorem 4.1 (zero-padding transcendence). *(Proven, with a finite-state component as stated below.) There is an explicit family of full-variable rules, one at each $d \geq 5$, related by a coefficient-preserving lifting, under which* 60714 *is a fixed point at every $d \geq 5$ (Proven), universal at $d = 5$ and $d = 6$ (Computed, exhaustive), and universal on $A_d \setminus E_d$ at every $d \geq 7$, where A_d is the admissible* *set and E_d is the escape class of Definition 4.6, whose orbits collapse to* 0. *The reaching-time bound underlying the induction is proven algebraically for odd $d \geq 15$ and even $d \geq 18$ (Appendix E) and closed by complete enumeration for $7 \leq d \leq 16$ — about* 25 *million multisets, re-run in full for this paper.*

The construction and the proof skeleton follow; the point of including them is that every step leans on the zeros.

The ladders. The native rule has $c^{(5)} = (9900, 9, 90, -9000, -999)$ and acts on sorted form $(7, 6, 4, 1, 0)$. For $d \geq 7$ odd, define $c^{(d)}$ by appending to $c^{(d-2)}$ the zero-sum pair $(+9 \cdot 10^{d-2}, -9 \cdot 10^{d-2})$ at the two new bottom ranks. The even ladder lifts the same way from the root

$$c^{(6)} = (9900,\ 9,\ 90,\ -9000,\ 99000,\ -99999),$$

the native -999 split across the two zero positions, $99000 - 99999 = -999$. Both ladders preserve the four coefficients at 60714's nonzero digits verbatim.

Proposition 4.2 (fixedness is free). *(Proven.) Every rule on either ladder fixes* 60714.

Proof. The sorted form of the padded fixed point is $(7, 6, 4, 1, 0, \ldots, 0)$. All appended (and split) coefficients sit at ranks holding zero digits, so $\sum_i c_i f_i = 9900 \cdot 7 + 9 \cdot 6 + 90 \cdot 4 - 9000 \cdot 1 = 60714$ at every length. ∎

This was verified mechanically at every $d \leq 20$ and at $d = 100$ (Computed). The content of Theorem 4.1 is never the fixed-point equation; it is the basin, and the basin argument has three moving parts.

Lemma 4.3 (absorbing set). *(Proven.) Let T_d be the admissible multisets whose sorted form ends in two zeros. For $d \geq 7$ on either ladder and $n \in T_d$,*

$$K^{(d)}(n) = K^{(d-2)}((s_0, \ldots, s_{d-3})), \quad \text{and} \quad K^{(d)}(n) \in T_d.$$

Proof. The appended pair multiplies $s_{d-2} = s_{d-1} = 0$ and vanishes; what remains is the rule two rungs down on the first $d-2$ sorted digits, since those coefficients are copied verbatim. The output is below 10^{d-2}, so its d-digit padding has at least two zeros, which sort to the tail. ∎

Lemma 4.4 (core positivity). *(Proven.) For every sorted-descending s, the native part* $\mathrm{core}(s) = 9900s_0 + 9s_1 + 90s_2 - 9000s_3 - 999s_4$ *satisfies* $0 \leq \mathrm{core}(s) \leq 89{,}991 < 10^5$.

Proof. The upper bound drops the negative terms. For the lower bound: if $s_3 = s_4 = 0$ all terms are nonnegative. If $s_3 \geq 1, s_4 = 0$, then $\mathrm{core} = 9000(s_0 - s_3) + 900s_0 + 9s_1 + 90s_2 \geq 0$. If $s_3, s_4 \geq 1$, use $s_1 \geq s_4$ and $s_2 \geq s_3$, then $s_3, s_4 \leq s_0$:

$$\mathrm{core}(s) \ \geq\ 9900s_0 + 9s_4 + 90s_3 - 9000s_3 - 999s_4 \ =\ 9900s_0 - 8910s_3 - 990s_4 \ \geq\ 9900(s_0 - s_3) \ \geq\ 0. \qquad \blacksquare$$

The even ladder's six-coefficient root obeys the same bounds one decimal place higher: expanding $99000s_4 - 99999s_5 = -999s_4 + 99999(s_4 - s_5)$ gives $\mathrm{core}_6(s) = \mathrm{core}(s_0, \ldots, s_4) + 99999\,(s_4 - s_5)$, and since $0 \leq s_4 - s_5 \leq 9$, $0 \leq \mathrm{core}_6(s) < 10^6$. (The additive estimate $89{,}991 + 9 \cdot 99{,}999 = 989{,}982$ over-counts, its two extremes being incompatible; the exact maximum, computed in Appendix E.2, is 899,991.)

Lemma 4.5 (zeros are produced). *(Proven.) Each appended pair contributes $9 \cdot 10^{e_k}\, \delta_k$ with $\delta_k \geq 0$ a digit gap and the place e_k two higher than the previous pair's; the δ_k are disjoint adjacent* *differences in a sorted string, so $\sum_k \delta_k \leq 9$. By Lemma 4.4 and its even-ladder extension the absolute value is never active and the output decomposes, without carries, into two-digit blocks (the digits of $9\delta_k$) sitting above a five-digit core (odd ladder) or six-digit core (even ladder). A block contributes two zero digits if $\delta_k = 0$, one if $\delta_k = 1$, none if $\delta_k \geq 2$; since $2\,\#\{\delta_k \geq 2\} + \#\{\delta_k = 1\} \leq \sum_k \delta_k \leq 9$, the output has at least $2M - 9$ zeros, M the number of pairs. For $M \geq 6$ — that is, $d \geq 17$ on the odd ladder, $d \geq 18$ on the even — this is at least two zeros: every orbit enters T_d in one step. A refined case analysis lowers the odd-ladder threshold to $d \geq 15$; it and the even ladder's full argument are in Appendix E.*

For $7 \leq d \leq 16$ the same conclusion (entry into T_d within at most eight steps) holds by complete enumeration over all admissible multisets, every length re-run in full for this paper; the table is in Appendix E. One detail of the induction deserves its own lemma. When Lemma 4.3 drops an orbit to length $d-2$, the projected multiset can fail admissibility there: a repdigit projection is block-aligned and belongs to the escape class, while a near-repdigit projection recovers admissibility within two further steps (Appendix E, projection lemma). Induction along each ladder then proves Theorem 4.1: orbits fall into T_d, where Lemma 4.3 reduces the length by two, the projection lemma hands the reduced orbit to the inductive hypothesis, and the roots $d = 5, 6$ are universal by the census. The ladder rules themselves are written out through $d = 20$, with the $d = 100$ instance, in Appendix F.

Definition 4.6 (escape class). The *block-aligned* multisets — constant on the native block and on each appended pair — are sent to 0 in one step, because the native coefficients sum to zero and each pair sees equal digits. E_d is the backward orbit of this set. At $d = 5, 6$ it is empty (the censuses show universality there, with no escape). At $d = 7$ the one-step part is the 45 multisets (x^5, y^2), $x > y$, and the full classification run for this paper found $|E_7| = 81$ of 11,340 admissible multisets collapsing to 0 with all other 11,259 reaching 60714; at $d = 8$, $|E_8| = 137$ of 24,210 (Computed). The one-step part has a closed form: the admissible block-aligned multisets number $\binom{9+k}{k} - 10$, k the number of blocks, which is 45 at both $d = 7$ and $d = 8$ — confirmed by direct count of the inputs with $K(n) = 0$ at each (Computed). The classical map has the same structure at its own small lengths — the multiples of 1111 at $d = 4$ collapse to 0 — so the escape class is a continuation of a classical phenomenon, not a new defect.

The contrast: 6174 does not climb. The classical constant admits the same kind of coefficient-preserving lifting — keep $(999, 90, -90, -999)$ on the nonzero digits of $(7, 6, 4, 1, 0, \ldots)$, append zero-sum pairs — and it never regains universality.

Theorem 4.7. *(Computed.) Padded* 6174 *has no full-variable fixed-point rule at all at $d = 5$ (none of the* 5,280 *rules fixes it — an algebraic obstruction, verified exhaustively here). At $d = 6$ exactly four full-variable rules fix it and the best basin is* 0.9686 *of the* 4,905 *admissible multisets (verified exhaustively here). Under the lifting family at $d = 7, 8, 9$ the best basins are* 0.9897, 0.9981, 0.9991 *[S], with the non-reaching inputs at $d = 8, 9$ being exactly the* 45 *multisets (X^4, Y^4), $X > Y \geq 0$, each collapsing to* 0 *[S].*

The structural diagnosis is short. 60714's native rule lands its largest coefficient, -999, on its zero digit: the zero positions, where lifting must do its work, already carry the rule's weight — which is also exactly why 60714 is degenerate at its native length (Theorem 3.2), the weight resting on a zero being precisely what the effective rank cannot count — and the absorbing set of Lemma 4.3 catches the escape candidates. 6174 has no zero digit at its native length, so its lifts pad with zeros *outside* the native coefficients; the (X^4, Y^4) inputs never meet the absorbing structure and survive as a permanent escape class. Where a fixed point keeps its zeros relative to its coefficients decides its fate across dimension. That is the lesson of the chain that works, and the standing problem of the chain that follows is that it has no zeros at all.

5 Lifting by duplication: folding

Let $M_m = \{1, 4, 6, 7\}^m$, a multiset of length $d = 4m$ with sorted form $\mathbf{s}_m = (7^m, 6^m, 4^m, 1^m)$. The duplication question asks for universal fixed points with this digit content at every m. The right algebraic tool is an operation on rules.

Definition 5.1 (folding). *Let $c = (c_0, \ldots, c_{d_0-1})$ be the coefficient vector of a rule at length d_0 and let $k \geq 1$. The k-fold $c^{\otimes k}$ at length kd_0 assigns to rank $ki + j$ ($0 \leq i < d_0$, $0 \leq j < k$) the coefficient $c_i \cdot 10^{d_0(k-1-j)}$.*

Proposition 5.2. *(Proven.) (a) $c^{\otimes k}$ is a valid rule (realizable by a permutation pair), full-variable if c is. (b) For every multiset X at length d_0, writing X^k for the multiset with each multiplicity multiplied by k,*

$$T^{\otimes k}(X^k) = \big(T(X)\big)^k :$$

the k-fold states are closed under the folded rule, and the folded dynamics on them is conjugate to the base dynamics. (c) If $K(W) = W$ then the k-fold rule fixes the concatenation $W^{(k)} = W \cdot \sum_{j<k} 10^{jd_0}$; if the base rule is universal, every k-fold state reaches $W^{(k)}$.

Proof. (a) If $c_i = 10^{a_i} - 10^{b_i}$ with $\{a_i\}$ and $\{b_i\}$ each a complete residue list $\{0, \ldots, d_0 - 1\}$, then the folded coefficients are $10^{d_0(k-1-j)+a_i} - 10^{d_0(k-1-j)+b_i}$, and $\{d_0t + a_i\}$, $\{d_0t + b_i\}$ each enumerate $\{0, \ldots, kd_0 - 1\}$ exactly once. (b) The sorted form of X^k repeats each sorted digit of X at k consecutive ranks, so ranks $ki, \ldots, ki + k - 1$ all hold s_i, and

$$\sum_{i,j} c_i 10^{d_0(k-1-j)} s_i = R_k \cdot \sum_i c_i s_i, \qquad R_k = \textstyle\sum_{j<k} 10^{jd_0}.$$

Taking absolute values, $K^{\otimes k}(X^k) = R_k \cdot K(X)$, and since $K(X) < 10^{d_0}$ — it is a difference of two d_0-digit numbers — multiplication by R_k concatenates k copies of its padded digit string without carries. The digit multiset of the output is therefore $(T(X))^k$. (c) Immediate from (b). ■

Folding explains the duplication chain's classical landmarks at a stroke.

Corollary 5.3. *(Proven.) The m-fold of the classical rule — the* interleaved *rule, with coefficients $999 \cdot 10^{4(m-1-j)}, 90 \cdot 10^{4(m-1-j)}, -90 \cdot 10^{4(m-1-j)}, -999 \cdot 10^{4(m-1-j)}$ on the four rank-blocks — fixes $F_m = 6174 \cdot R_m$, the string 6174 written m times, at every $m \geq 1$. Its restriction to m-fold states is conjugate to the classical four-digit map, so all of them lie in F_m's basin.*

So fixed points with the duplicated digit content exist at every m, provably. What the algebra does not control is everything off the folded subspace, and there the family comes apart.

Computed 5.4 (the interleaved basins). *Over all non-repdigit multisets (the duplication-chain convention), the basin of F_m under the interleaved rule is*

m	d	states	basin of F_m
1	4	705	100%
2	8	24,300	100%
3	12	293,920	14.86%

m	d	states	basin of F_m
4	16	2,042,965	99.69%
5	20	10,014,995	9.52%

All five rows were enumerated in full for this paper. At $m = 3$ the rule has exactly two fixed points — 617461746174 and 535549955994, the latter with a basin of 100 multisets — and 250,148 of the 293,920 states end in cycles.

The sequence $100, 100, 14.9, 99.7, 9.5$ is the family's character witness: one algebraically uniform construction, basins scattered across the unit interval with no trend. The second fixed point at $m = 3$ also matters for the record — it shows coefficient-positivity conditions (the rule is monotone in the sense below) do not force fixed-point uniqueness, a hypothesis the project's earlier notes briefly entertained and later refuted. It is also the first sighting of the structure that recurs at every failure of the duplication chain. The competitor 535549955994 is a relative of the *anti-Kaprekar* number 549945 — the rival three-pair constant built from the digits 4, 5, 9 — and the same content shadows the chain elsewhere: the period-two cycle that breaks the classical double-double (Computed 6.2) and the incoherent fixed point 655444440000 that breaks the $k = 3$ fold (Computed 6.7) are both heavy in 4, 5, and 9. The shadow is a family rather than a single number — its members surface as fixed points at some multiplicities and as cycles at others, and they do not stabilize across m — and that refusal to stabilize is precisely what blocks an induction that would exclude them (Section 7).

For the rest of the section, restrict to the rules adapted to M_m. Call a rule *pair-symmetric* if there are bijections between its seven-rank and one-rank coefficient blocks, and between its six-rank and four-rank blocks, under which corresponding coefficients are negatives. Write $S_7 = \sum_{7\text{-block}} c_i$ and $S_6 = \sum_{6\text{-block}} c_i$, and let $C_j = c_0 + \cdots + c_j$ be the partial sums, so that for any sorted input, using $\sum c_i = 0$,

$$\sum_i c_i s_i \;=\; \sum_{j=0}^{d-2} C_j\, g_j, \qquad g_j = s_j - s_{j+1} \geq 0.$$

Proposition 5.5. *(Proven.) For a pair-symmetric rule at $d = 4m$: (a) $C_{m-1} = C_{3m-1} = S_7$ and $C_{2m-1} = S_7 + S_6$; (b) the value on the sorted form $\mathbf{s}_m$, whose only nonzero gaps are $g_{m-1} = 1$, $g_{2m-1} = 2$, $g_{3m-1} = 3$, is*

$$V \;=\; 6S_7 + 2S_6 \;=\; 4C_{m-1} + 2C_{2m-1};$$

(c) the rule fixes an arrangement of M_m if and only if $|V|$ has digit multiset M_m, a condition depending only on the partition of positions into the four blocks, not on the within-block coefficient assignment; (d) the interleaved rule is monotone — $C_j \geq 0$ for all j — at every m.

Proof. (a) Pair-symmetry gives $S_1 = -S_7$ and $S_4 = -S_6$, so $C_{3m-1} = S_7 + S_6 + S_4 = S_7 = C_{m-1}$. (b) Substituting into the gap form, $V = C_{m-1} + 2C_{2m-1} + 3C_{3m-1} = S_7 + 2(S_7 + S_6) + 3S_7 = 6S_7 + 2S_6$; directly, $V = 7S_7 + 6S_6 + 4S_4 + 1 \cdot S_1 = 6S_7 + 2S_6$. (c) The multiset map sends M_m to the digit multiset of $|V|$; if that is M_m the orbit of M_m is fixed and the fixed integer is $|V|$. Both S_7 and S_6 are sums of $10^a - 10^b$ over the block's position pairings, hence depend only on the position sets. (d) For the interleaved rule the partial sums climb through the seven- and six-blocks; inside the four-block, $C = 999R_m + 90R_m - 90\sum_{k \geq m-t} 10^{4k} \geq 999R_m \geq 0$ after t steps; inside the one-block, $C = 999(R_m - \sum_{k \geq m-t} 10^{4k}) \geq 0$. ■

Part (c) reduces the *fixing* question to explicit arithmetic over partitions — at $m = 2$, for instance, exactly 36 of the 2,520 partitions fix an arrangement of M_2 (Computed) — and part (d), with Corollary 5.3, says the algebraic requirements of the duplication conjecture (a fixing, monotone, full-variable rule at every m) are met uniformly. By Lemma 2.4, everything open lives in two dynamical conditions: no second fixed point, no cycle.

Computed 5.6 (the universal landscape, $m \leq 6$). *Universal fixed points of M_m exist at $m = 1, \ldots, 6$, and (on the alphabet-complete-plus-large-sample standard, not full enumeration) at $m = 7$. The counts of universal arrangements are* 2, 465 *(*481 *under the strict near-repdigit convention),* 46, $\geq$ 313, $\geq$ 1, $\geq$ 1 *[S]. Within the pair-symmetric class at $m = 2$, this paper's exhaustive enumeration found* 312 *universal rules concentrated on* 12 *arrangements (Appendix C). The clean block point $F_m = 6174$ repeated is not universal under the interleaved rule for any $m \geq 3$ (its basins are the table above), and the $m = 3$ universals found are scrambled [S]; but block decomposition is not itself the obstruction, since at $m = 4$ and $m = 5$ folding makes block-decomposed arrangements universal —* 6174 *and* 1746 *each repeated m times among them (Computed 6.4). The following full-basin verifications were performed for this paper, each over the complete non-repdigit multiset space:*

m	d	states	a verified universal	rule
3	12	293,920	666141417774	the project's $m = 3$ rule, re-verified
4	16	2,042,965	1746174617461746	$(9, -900, 900, -9)^{\otimes 4}$
5	20	10,014,995	17461746174617461746	$(9, -900, 900, -9)^{\otimes 5}$
6	24	38,567,090	666141417774 written twice	the $m = 3$ rule, 2-folded

*The $d = 20$ row is, to the author's knowledge, the first complete-enumeration universality verification at $m = 5$; the project's previous $m = 5$ witness (*14617461774617461746*) rested on a necessary-condition proxy [S]. The $d = 24$ row gives a second verified universal arrangement at $m = 6$ alongside the project's* 666174141466617777741414 *[S].*

At $m = 7$ ($d = 28$) the clean repeat — 6174 written seven times — is a universal fixed point on the standard the $m = 5$ witness held before its enumeration: alphabet-complete over all 4,491 multisets on $\{1, 4, 6, 7\}$ and clean over a 500,000-input random sample, under an explicit monotone pair-symmetric rule (Appendix D). Full enumeration of the $\sim 1.2 \times 10^8$ admissible multisets at $d = 28$ is beyond exhaustive reach here, so this is evidence, not a complete-enumeration proof. It is the first prime multiplicity the doubling tower cannot reach; the naive fold leaks into a 4995/5355 cycle that the right within-block reordering removes.

Conjecture 5.7. *M_m carries a universal full-variable fixed point at every $m \geq 1$.*

The evidence now includes six consecutive multiplicities with full-basin witnesses (and a seventh, $m = 7$, on the large-sample standard), and witness-count estimates that grow like $3 \cdot 10^3$, $2 \cdot 10^7$, $3 \cdot 10^8$ at $m = 3, 4, 5$ [S, sampling estimates calibrated against the exact $m = 3$ enumeration]. What it does not include is any uniform description: the universal arrangements for $m \geq 3$ are scrambled, and the structural features that characterize them at one multiplicity provably fail at the next (Section 7). The table's most interesting column is the last one, and it is the subject of the next section: four of the six witnesses are folds.

6 When folding preserves universality

Proposition 5.2 transports the fixed point along $m \mapsto km$ for free; it says nothing about the basin off the folded subspace. The experiments reported here measure exactly that. All "universal" verdicts below are complete enumerations; all "fails" verdicts are definitive, because the witness of failure is an admissible input (a multiset over $\{1, 4, 6, 7\}$) that demonstrably does not reach the fixed point. The rules used are written out in Appendix D.

Computed 6.1 (first doubling: everything survives at $d = 4$). *The* 2*-folds of all eight universal full-variable rules at* $d = 4$ *(Theorem 3.2) are universal at* $d = 8$*, each verified over all* 24,300 *multisets. In particular the* 2*-fold of the classical rule is the interleaved* $m = 2$ *rule, whose universality is the* $m = 2$ *row of Computed 5.4. This is a property of the four-digit bases, not a law: the* 2*-fold of* 60714*'s native five-digit rule fixes* 6071460714 *at* $d = 10$ *with basin* 53.34% *of the* 92,368 *non-repdigit multisets — the chain that climbs perfectly by zero-padding does not climb by doubling.*

Computed 6.2 (second doubling: selection). *Iterating the doubling, exactly one of the four* $d = 4$ *fixed points survives: the* 2*-fold of the* 2*-fold of* $(9, -900, 900, -9)$ *— the* 1746 *rule — is universal at* $d = 16$ *(*2,042,965 *multisets), while the doubled doubles of the* 2538*,* 5382*, and* 6174 *rules all fail, each trapping admissible* $\{1, 4, 6, 7\}$*-multisets. For the classical rule the obstruction is a period-two cycle through the multisets of* 5355517553558172 *and* 4995599449954176*, which captures* 14 *of the* 965 *alphabet multisets.*

Computed 6.3 (the 1746 fold spectrum). *For the* 1746 *rule* $c = (9, -900, 900, -9)$ *and* $2 \le k \le 8$*:*

k	d	verdict	basis
2	8	**universal**	complete enumeration, 24,300
3	12	fails (15.48%)	complete enumeration
4	16	**universal**	complete enumeration, 2,042,965
5	20	**universal**	complete enumeration, 10,014,995
6	24	fails	alphabet witness (523/2921 reach F)
7	28	fails	alphabet witness
8	32	fails	alphabet witness (6533/6541)

The $k = 4$ *entry is the direct* 4*-fold; the iterated double-double — a different within-block assignment on the same partition — is also universal at* $d = 16$ *(Computed 6.2). The* 5*-folds of the other three* $d = 4$ *rules fail at* $k = 5$*; the* 3*-folds of all four fail. Mixed towers also fail where tested (*3*-fold of the* 2*-fold at* $d = 24$*: fails). The iterated double-double-double of the* 1746 *rule at* $d = 32$ *— which differs from the direct* 8*-fold only in its within-block coefficient assignment — fixes* 1746 *written eight times, sends all* 6,541 *alphabet multisets and* 200,000 *random admissible multisets to it, and exceeds exhaustive reach at* $\binom{41}{9} \approx 3.5 \times 10^8$ *states (Evidence).*

Computed 6.4 (folding from $m = 2$: survival is common). *Of the* 312 *universal pair-symmetric rules at* $m = 2$*, exactly* 144 *— spanning nine of the twelve universal arrangements*

— *have universal* 2*-folds at* $m = 4$*, every one verified by complete enumeration over all* 2,042,965 *multisets. Among them are sixteen rules whose folds fix* $F_4 = 6174$ *written four times: the block fixed point that the interleaved rule fails to make universal (*99.69%*, Computed 5.4) is made universal by other doublings of the same partition, which explains, in folding terms, the project's observation that* F_4 *is universal only under scrambled within-block orderings [S]. The alphabet test was a perfect filter here:* 144 *of* 312 *doubled rules passed it, and all* 144 *proved universal.*

Computed 6.5 (folding from $m = 3$**).** *The* 2*-fold of the project's verified* $m = 3$ *universal rule is universal at* $m = 6$*: all* 38,567,090 *non-repdigit multisets at* $d = 24$ *reach* 666141417774666141417774*.*

Three things follow from this body of computation, and a fourth is suggested.

First, the duplication chain *does* have a working lifting — in instances. The project's earlier searches had tested and rejected a position-insertion lifting $m \to m + 1$ (best basin 3.07% across all natural variants [S]) and a family of m-periodic constructions (each universal at some multiplicities and provably cyclic at others [S]); the conclusion drawn was that universality does not transport. Folding shows the conclusion was too broad. The map $m \to km$ transports the entire algebraic package by Proposition 5.2, and the dynamics follows it in verified cases substantial enough to populate Computed 5.6's table: a single four-digit rule covers $m \in \{1, 2, 4, 5\}$ and a single twelve-digit rule covers $\{3, 6\}$.

Second, the survival pattern is irregular — $k = 2, 4, 5$ work from the 1746 root and $k = 3, 6, 7, 8$ do not, the direct 8-fold fails while the iterated doubling passes every feasible test, and the within-block assignment (the only difference between those two) is decisive. This is the duplication chain's recurring signature, now visible inside a single rule's fold family.

Third, doubling-survival is not rare: eight of eight rules from $m = 1$, 144 of 312 from $m = 2$, one of one tested from $m = 3$. At the only two doubling steps where the survival fraction could be measured exhaustively it is 100% and 46%, which makes the truth of Conjecture 5.7 look like the generic outcome of an abundant mechanism rather than a sequence of coincidences — the strongest structural evidence the conjecture currently has. What *is* selective is iterated survival along a fixed tower: of the four $d = 4$ roots, only the 1746 rule's tower passes the second doubling, and it then passes everything testable at the third. The selection is precisely parallel to 60714's — one survivor of an exhaustive cross-dimensional test — and it nominates a single candidate for the duplication chain's L5 occupant.

Conjecture 6.6 (dyadic tower). *The iterated* 2*-fold of* $(9, -900, 900, -9)$ *is universal at* $d = 2^{j+2}$ *for every* $j \geq 0$ *— equivalently,* 1746 *written* 2^j *times is a full universal fixed point at every dyadic multiplicity.*

This is the sharpest available subconjecture of Conjecture 5.7: a single explicit self-similar family, fixedness proven at every level, universality fully verified at $j \leq 2$ and supported by all feasible evidence at $j = 3$. It is worth stating why a proof might be less hopeless here than for the general conjecture. The earlier no-go evidence [S] rules out *m-periodic* uniform constructions — rules whose position structure repeats with the block — by exhibiting cycles at specific multiplicities. The folded towers are not periodic; their coefficient structure is self-similar across scales, with the base rule's geometry reproduced at every dyadic level, and the no-go arguments do not apply to them. What is missing is any analogue of Lemma 4.3: a closed set that the folded dynamics provably enters in bounded time and on which the multiplicity provably drops. The k-fold states are closed and reduce km to m (Proposition 5.2(b)), but nothing forces an orbit *into* them — they are an atoll, not a drain. Finding a fold-adapted absorbing structure, playing the role zeros play in Section 4, is in the author's view the most concrete open *proof* problem this subject now offers.

That problem now has a candidate answer, found by asking what the fold does to states *off* the folded subspace. At length kd_0, rank $ki+j$ carries the coefficient $c_i\,10^{\,d_0(k-1-j)}$, so for any sorted state x — folded or not —

$$K^{\otimes k}(x) \;=\; \Big|\sum_{j<k} 10^{\,d_0(k-1-j)}\;V\big(x^{(j)}\big)\Big|, \qquad x^{(j)} = (x_{ki+j})_{i<d_0},$$

where $V(t) = \sum_i c_i t_i$ is the base linear form and the *slices* $x^{(j)}$ are themselves sorted, the j-th entry of each rank-block. For the 1746 rule, $|V| \le 8019 < 10^4$ on sorted tuples, so the slice values occupy disjoint decimal blocks of the output. Call x *coherent* when its nonzero slice values share a sign. On a coherent state the absolute value distributes across blocks: the image integer is the slice values written blockwise, its digit multiset is the disjoint union of theirs, and the entire future of the orbit therefore depends only on the tuple $\big(|V(x^{(0)})|, \ldots, |V(x^{(k-1)})|\big)$. The folded states are the coherent states with all slices equal; coherence is the enlargement that turns out to drain.

Computed 6.7 (coherence discriminates survival). *Every non-repdigit state becomes coherent within* 3 *steps at* $k = 2$ *(all* 24,300 *multisets) and within* 5 *steps at* $k = 4$ *(all* 2,042,965*); a* 300,000-*multiset random sample at* $k = 5$*,* $d = 20$ *enters within* 4 *steps with no exceptions. At* $k = 3$ *— the failing fold — coherence is not reached from everywhere: the multiset* $\{9^8, 5, 4^3\}$ *falls into* 655444440000*, a second fixed point of the* 3*-fold rule, incoherent with slice signs* $(+,+,-)$*. The fold's failure mode is an incoherent fixed point.*

Computed 6.8 (the pair quotient funnels at both tower levels). *On coherent states the factorization through slice values is exact — zero violations over the full enumerations. At tower level one (*$k = 2$*,* $d = 8$*) the reachable pair system has* 1,464 *states; the* 1,202 *whose forward orbits never exit coherence form a funnel: unique fixed pair* $(1746, 1746)$*, no cycles. At level two (*$d = 16$*) the tower element* $(c^{\otimes 2})^{\otimes 2}$ *is itself universal — a complete enumeration over all* 2,042,965 *multisets, new here alongside Computed 6.2 — its states become coherent for the level-two pairing within* 3 *steps (against* 5 *for the flat four-slice decomposition: the nested pairing is the dynamically natural one), and its pair system of* 235,212 *states funnels to the unique fixed pair* $(17461746,\ 17461746)$ *with no cycles.*

The recursion between levels is syntactic. Writing V_j for the level-j form, the level above evaluates the level below on the two parity slices:

$$V_{j+1}(t) \;=\; 10^{\,d_j}\,V_j(t_{\text{even}}) + V_j(t_{\text{odd}}),$$

which is how the analogy that has run through this paper closes. The classical four-digit map reduces to a 54-state gap system funnelling to $(6, 2)$ (Section 7); the coherent fold dynamics reduces, one level up, to a 1,464-state pair system funnelling to $(1746, 1746)$; the level above that, to a pair system over the level-one values funnelling to the level-one fixed point doubled. The funnel boundary is no simpler than the base case's — no inequality or congruence tested separates the 1,202 good pairs from the 262 exiting ones, the certificate-cost obstruction of Section 7 casting its shadow upward — but the reduction itself is exact, and it converts Conjecture 6.6 into two uniformities: bounded entry into coherence at every level, and the funnel property of every level's pair system. Both are verified at the two levels within enumeration's reach; both are open above, where the pair systems outgrow it. Section 7 states the sharpened problem.

A remark on what folding cannot do even in principle. A fold realizes multiplicity m only from a universal base at some m_0 properly dividing m, so folds from the verified bases $m_0 \le 6$ can at best reach the m whose least prime factor is at most 5 — and reach them only when the particular fold survives, which Computed 6.3 shows is not for free. The multiplicities with least prime factor ≥ 7, beginning with $m = 7, 11, 13$, need witnesses of their own at the bottom of their divisor chains. Folding reorganizes the conjecture around its prime levels; it does not exhaust it.

7 The cost of a certificate

Every universality statement in this paper is, by Lemma 2.4, the statement that a finite functional graph has one root and no cycles. Such statements always have proofs — trace the graph — and the question that separates Section 4 from Section 6 is whether they have proofs *smaller than the graph*. For the zero-padding chain they do: Lemmas 4.3–4.5 are a certificate of total size independent of d. This section measures the same question at the base of the duplication chain and finds the answer sharply different.

The classical four-digit map factors through two coordinates. For sorted digits $a \geq b \geq c \geq e$, the map computes $999(a-e) + 90(b-c)$, so with $p = a - e \in \{1, \dots, 9\}$ and $q = b - c \in \{0, \dots, p\}$ the dynamics lives on 54 states, with unique fixed point $(6, 2)$ (this is 6174: $p = 7 - 1$, $q = 6 - 4$). The system is acyclic with maximal reaching time 6, and its iterated images shrink as $54, 20, 14, 10, 7, 4, 1$ (Computed). A *polynomial Lyapunov function of degree D* is a $\Phi \in \mathbb{R}[p, q]$, $\deg \Phi \leq D$, with $\Phi(\text{next}) < \Phi(\text{current})$ at all 53 non-fixed states; its existence is a linear program in the $\binom{D+2}{2}$ coefficients.

Theorem 7.1 (the Lyapunov degree of Kaprekar's theorem is exactly seven). *(Proven.) No polynomial Lyapunov function of degree ≤ 6 exists for the four-digit gap system: for each $D = 2, \dots, 6$ there is an exact rational Farkas certificate — a nonnegative rational combination of the 53 decrease constraints summing to a contradiction — verified in integer arithmetic (the $D = 4$ certificate is supported on 15 states; Appendix B). A Lyapunov function of degree 7 exists: an explicit rational polynomial, verified in exact arithmetic to decrease strictly along all 53 transitions.*

Two readings of this theorem are wrong and one is right. It does not say the convergence is unprovable — it is provable by a 54-state enumeration, and indeed by a 36-coefficient polynomial. It also does not say (as a first pass over this project's data once suggested) that certificates exist only at the trivial interpolation threshold, degree 9, where polynomials span all functions on the state set and the reaching time itself becomes a vacuous witness; degree 7 is strictly below that threshold, so a genuinely non-tautological algebraic certificate exists. What the theorem says is quantitative: the cheapest polynomial witness of the seventy-five-year-old fact costs 36 coefficients on a 54-point system. Nothing about it is light, and nothing about it is structural — the certificate is a found object, not an explanation, and there is no visible way to write it as the first member of a family indexed by m.

That is the correct statement of the base-case obstruction, and it propagates. Each of the standard strategies for Conjecture 5.7 was pursued seriously in this project and failed for a reason now identifiable with the base case's lack of transportable structure; the table records the sharp form of each failure (all [S], from the project's verified computational log; the basin and enumeration figures quoted were re-verified here where marked).

strategy	sharp obstruction
uniform / periodic construction	the one clean recipe universal at $m = 3, 4$ fails at $m = 2$ and $m = 5$ (period-8 and period-16 cycles); position-insertion lifting $m \to m+1$: best basin 3.07%
induction on excluded cycles	obstruction cycles do not stabilize: the $m = 5$ cycle families do not occur at $m = 3, 4$; each multiplicity grows new ones
structural characterization of universal rules	the property characterizing universal partitions at $m = 3, 4$ (seven-block at the top) yields zero universals at $m = 5$ in an 11.3-million-sample search; the verified $m = 5$ winner has a different shape
Lyapunov / monovariant	LP-infeasible over digit-count and quadratic feature classes at $d = 16$; at the base, Theorem 7.1
modular invariant	no modulus in $2, \ldots, 300$ separates the basin from the cycles; mod 9 (Lemma 2.2) is universal and hence empty
union bound / local lemma over cycles	the dominant cycle alone captures $\sim$67% of candidate rules — far too dense
counting lower bound	the universal fraction is erratic (6.94% exact at $m = 3$, $\sim$28% at $m = 4$, $\sim$0.04% at $m = 5$); only the absolute count grows, and no provable lower bound for it is known

The common root is uniformity: universal rules exist in abundance at every tested m, but the *description* of where they sit changes with m, which simultaneously defeats construction, induction, characterization, and any local-to-global probabilistic argument. A workable proof must either find structure that is genuinely scale-free — the dyadic towers of Section 6 are the first candidate family of that kind to survive testing — or dispense with description entirely and prove existence by a counting argument robust to wild fluctuation in the universal fraction. Both are research programs, not techniques.

It is worth closing the circle with Section 4. The zero-padding proof never confronts any of this, because it never proves convergence at a fixed length at all: it proves a *reduction between lengths* (Lemma 4.3) and pays a bounded, d-independent cost to reach the reducing set (Lemma 4.5). The duplication chain now has both pieces in verified form — the reduction is the coherent factorization of Computed 6.8, the reaching is the bounded coherence entry of Computed 6.7 — but each is established by enumeration at two tower levels, not by an argument uniform in the level. The precise frontier, sharpened by Section 6's findings:

Open Problem 7.2. *For the dyadic tower of the* 1746 *rule, prove uniformly in the level j the two statements verified by complete enumeration at $j = 1, 2$: (E) every non-repdigit state becomes coherent within a bounded number of steps; (F) the coherent pair quotient has a unique fixed pair — the level below's fixed point doubled — and no cycles. By the exact factorization of Computed 6.8, (E) and (F) together imply Conjecture 6.6. Lemma (E) is a carry-and-block argument one* *level above Lemma 4.5 and appears attackable; lemma (F) must exploit the syntactic recursion $V_{j+1} = 10^{d_j} V_j \oplus V_j$, since the pair systems grow past enumeration and their funnel boundaries admit no simple description.*

Open Problem 7.3. *Prove any lower bound on the number of universal rules for M_m that is positive for infinitely many m.*

8 Methods and provenance

All computations are over digit multisets; a rule's orbit depends only on the sorted digit vector, which is what makes complete enumeration feasible — $\binom{d+9}{9}$ states minus exclusions, against 10^d integers. Conventions: the censuses of Section 3 and the 60714 chain exclude repdigits and near-repdigits (615, 1,902, 4,905, 11,340, 24,210 admissible multisets at $d = 4, \ldots, 8$); the duplication chain excludes repdigits only (24,300; 293,920; 2,042,965; 10,014,995; 38,567,090 at $d = 8, 12, 16, 20, 24$). Basins were computed by orbit tracing with terminal memoization; the $d = 20, 24$ enumerations use a combinatorial ranking of multisets into a flat byte array (at most $\binom{d+9}{9}$ bytes of state), which is what brings the previously "infeasible" $d = 20$ complete basin (Computed 5.6) to under two minutes in an interpreted language. The Lyapunov LPs of Theorem 7.1 were solved with an exterior solver and then *re-verified in exact rational arithmetic*: the infeasibility certificates are nonnegative rational vectors checked to annihilate the constraint matrix over $\mathbb{Q}$, and the degree-7 function was rationalized and its 53 strict decreases checked exactly.

Results marked [S] are quoted from the project's computational log and were not re-executed for this paper; they comprise the 6174 lifting basins at $d = 7, 8, 9$, the exhaustive one-step closure confirmations at $d = 17, 18, 20$ cited in Appendix E, the duplication-chain counts at $m \geq 3$ and their witness-count estimates, the $m = 3$ complete pair-symmetric enumeration (3,000 universal rules among 43,200 monotone fixing rules), the obstruction-cycle and no-go analyses summarized in Section 7's table, and the $m = 6$ witness 666174141466617777741414. The $d \leq 16$ reaching-time enumeration behind Theorem 4.1 — about 25 million multisets — was re-run in full for this paper (Appendix E.3), as were the one-step escape counts at $d = 7, 8$ and the $d = 6$ census lists of Appendix A. Everything else — the full censuses at $d \leq 6$ with the collapse counts and the selection theorem, the 60714 ladder checks through $d = 100$, the Case-2 projection closed forms confirmed through $d = 401$, and full classifications at $d = 7, 8$, the five interleaved basins, the two-fixed-point inventory at $d = 12$, the $m = 2$ pair-symmetric enumeration, every fold verdict in Section 6 — including the 144 complete $d = 16$ basin enumerations of Computed 6.4 and the complete enumerations at $d = 8, 20, 24$ — the coherence-entry and quotient analyses of Computed 6.7–6.8 (complete enumerations at $d = 8, 12, 16$ including the tower element's universality, with the $k = 5$ entry bound from a 300,000-multiset random sample at $d = 20$), and both directions of Theorem 7.1 — was computed afresh for this paper from the engine, and the verification scripts accompany the manuscript.

Acknowledgments

The enumerations, searches, and verification harness were built and run in collaboration with Anthropic's Claude, used as a research instrument; several of the project's intermediate claims (a uniqueness hypothesis for monotone rules; a first reading of the Lyapunov threshold) were retracted or corrected when verification contradicted them, and the corrected statements appear above. The questions, the direction, and the responsibility for the results are the author's.

A The census lists

54, 3753, 12456, 14562, 15642, 16524, 16578, 16758, 17685, 18342, 21456,

21834, 24156, 24183, 24561, 28539, 37584, 37854, 38754, 41562, 41832, 42183,

43758, 43785, 43875, 45612, 45621, 53928, 58239, 60417, 60714, 65781, 67581.

All thirty-three were found by exhaustive enumeration of the 5,280 full-variable rules against the 1,902 admissible multisets. The four with $\mathrm{sv}_F < 5$ are exactly those whose sorted form contains a zero digit: 54 ($\mathrm{sv}_F = 2$), 3753 ($\mathrm{sv}_F = 4$), and the two arrangements 60417, 60714 of $\{0, 1, 4, 6, 7\}$ ($\mathrm{sv}_F = 4$ each). All four are degenerate universals — the five-digit analogues of the degenerate three-digit constants — and of the thirty-three only 60714 survives to $d = 6$ (Theorem 3.3).

At $d = 6$, the high-zero strata of the 506 — where cross-dimensional behavior concentrates — are small enough to list, and both lists below were checked against this paper's recomputed census. The eight fixed points with three zero digits are all arrangements of the single multiset $\{0, 0, 0, 2, 2, 5\}$:

252, 2520, 20025, 25200, 200025, 200250, 250200, 252000.

The fifty-three with two zero digits fall into eight multiset clusters, of sizes 32 ($\{9, 9, 8, 1, 0, 0\}$), 7 ($\{5, 5, 4, 4, 0, 0\}$), 5 ($\{9, 7, 1, 1, 0, 0\}$), 4 ($\{7, 6, 4, 1, 0, 0\}$ — the thread of Theorem 3.3: 60714, 146070, 170460, 607140), 2 ($\{7, 7, 3, 1, 0, 0\}$), and three singletons (9225, 67023, 52605). The full sorted list:

4545, 7191, 8919, 8991, 9189, 9225, 10899, 17019, 37017, 44505, 52605, 54450,

60714, 67023, 80919, 80991, 81099, 89019, 90819, 90918, 91809, 98091, 100899,

108099, 109809, 146070, 170460, 180909, 189009, 190089, 190809, 445005,

445050, 504450, 544500, 607140, 700191, 701901, 707130, 719100, 809091,

809109, 810909, 890091, 900891, 901890, 908091, 908109, 908190, 908901,

909081, 910089, 910809.

The remaining 445 fixed points (205 with no zero digit, 240 with one) are in the repository's census file.

B The degree-4 Farkas certificate

For $D = 4$ the Lyapunov LP has 15 unknowns and 53 constraints, one per non-fixed gap state (p, q), each of the form $w \cdot \big(\mathbf{m}(G(p, q)) - \mathbf{m}(p, q)\big) \le -1$ with $\mathbf{m}$ the monomial vector of degree ≤ 4 and G the gap map. The exact infeasibility certificate found and verified for Theorem 7.1 is a rational vector $y \ge 0$, supported on the fifteen states

$(1, 0)$, $(2, 0)$, $(4, 2)$, $(4, 3)$, $(5, 1)$, $(5, 3)$, $(5, 5)$, $(6, 0)$, $(6, 3)$, $(7, 1)$, $(7, 3)$, $(8, 0)$, $(9, 2)$, $(9, 6)$, $(9, 9)$,

with common denominator 10,885,528,810, satisfying $\sum_s y_s\big(\mathbf{m}(G(s)) - \mathbf{m}(s)\big) = 0$ exactly and $\sum_s y_s = 1$: any Φ of degree ≤ 4 would have to decrease along a nonnegative combination of transitions that exactly cancels, which is absurd. Certificates for $D = 2, 3, 5, 6$ have supports of sizes 6, 10, 21, 28 and were verified the same way. The degree-7 Lyapunov function has 36 rational coefficients and worst strict-decrease margin -0.979 after rationalization; its coefficient vector is in the accompanying repository.

C The pair-symmetric universal rules at $m = 2$

Exhaustive enumeration for this paper: 2,520 ordered partitions of the eight positions into the four blocks, of which 36 fix an arrangement of M_2 (Proposition 5.5(c)); 64 within-block coefficient assignments each; 312 of the resulting rules are universal over all 24,300 non-repdigit multisets, fixing 12 distinct arrangements with multiplicities

$$14617746,\ 17461746,\ 17746146\ (48\text{ rules each});\quad 61461774,\ 61746174,\ 61774614\ (32);$$

$$14177466,\ 17417466,\ 17741466\ (16);\quad 14661774,\ 17466174,\ 17746614\ (8).$$

Twenty-four of the 312 are monotone. The fold experiments of Computed 6.4 take this set as their base.

D Rules used in Section 6

The twelve-digit universal rule (Computed 5.6, $m = 3$; re-verified here over all 293,920 multisets) has coefficient vector

$$c = (99999999999,\ 9990000000,\ 999900000,\ 99999000,\ 999900,\ 9990,$$

$$-99999000,\ -999900,\ -9990,\ -99999999999,\ -9990000000,\ -999900000)$$

and fixed point 666141417774; it is pair-symmetric and is not itself a fold. Its 2-fold (Definition 5.1, $d_0 = 12$, $k = 2$) is the $d = 24$ rule of Computed 6.5. The k-folds of the 1746 rule are fully determined by Definition 5.1 from $c = (9, -900, 900, -9)$: rank $5i + j$ of the 5-fold at $d = 20$ carries $c_i \cdot 10^{16-4j}$, giving

$$(9 \cdot 10^{16}, 9 \cdot 10^{12}, 9 \cdot 10^{8}, 9 \cdot 10^{4}, 9,\ -900 \cdot 10^{16}, \ldots, -900,\ 900 \cdot 10^{16}, \ldots, 900,\ -9 \cdot 10^{16}, \ldots, -9),$$

with fixed point 1746 written five times. The iterated dyadic tower at $d = 32$ is $\left((c^{\otimes 2})^{\otimes 2}\right)^{\otimes 2}$, which differs from $c^{\otimes 8}$ in the within-block assignment of equal-magnitude coefficients to ranks; the former passes all feasible universality tests at $d = 32$, the latter fails the alphabet test (Computed 6.3).

In the letter notation of Section 2 the folds are visible to the eye. Label the sorted positions $a, b, c, \ldots$; at $d = 4m$ the ranks fall into four blocks of m equal digits, and copy j of base rank i is the j-th letter of block i. The 1746 rule and its first two surviving folds read:

m	d	π	σ	$\pi \cdot s - \sigma \cdot s$
1	4	`cbad`	`bcda`	$4671 - 6417$
2	8	`ecag fdbh`	`cega dfhb`	$46714671 - 64176417$
5	20	`kfap lgbq mhcr nids ojet`	`fkpa glqb hmrc insd jote`	$4671\cdots - 6417\cdots$

(Spaces are cosmetic.) Each four-letter group is the base pattern `cbad` (respectively `bcda`) written in the letters of successive copies, so the two rearrangements evaluate to 4671 and 6417 repeated m times, and the subtraction is the base subtraction $4671 - 6417 = -1746$ carried out blockwise: the pattern `cbad cbad` $\cdots -$ `bcda bcda` $\cdots$, in exact analogy with the classical `abcd` $-$ `dcba`. Every row was machine-checked to fix 1746 written m times (Computed). The same notation for the 60714 zero-padding ladder appears in the final appendix; there the rungs grow by prepended letter-pairs instead of repeated blocks, which is the two liftings' difference made typographic.

E The reaching-time bound in full

This appendix proves the reaching-time half of Theorem 4.1 at full rigor: every admissible orbit enters the absorbing set T_d within a bounded number of steps, the bound being one step for odd $d \geq 15$ and even $d \geq 18$, and at most eight steps for $7 \leq d \leq 16$ by complete enumeration. It closes with the projection lemma that the induction of Section 4 quietly uses.

The decomposition. By Lemmas 4.4 and 4.5 the absolute value in $K^{(d)}$ is never active, and the output splits without carries into a core plus disjoint two-digit blocks:

$$K^{(d)}(x) \;=\; \operatorname{core}(x) + \sum_{k=1}^{M} 9 \cdot 10^{e_k}\, \delta_k, \qquad \delta_k = x_{e_k-2} - x_{e_k-1} \in \{0, \ldots, 9\},$$

with M pairs at places e_k ascending by two. Writing $9\delta_k = 10u_k + v_k$, the block at places (e_k, e_k+1) carries digits (v_k, u_k): two zeros if $\delta_k = 0$, one zero ($u_k = 0$) if $\delta_k = 1$, none if $\delta_k \geq 2$, since $9\delta_k \in \{18, 27, \ldots, 81\}$ then has both digits nonzero. Each block value is at most $81 < 100$, so blocks never carry into one another; the core is below 10^5 (odd) or 10^6 (even) and at most one carry enters the lowest block, where the combined value $9\delta_1 + 1 \leq 82$ still occupies two digits.

E.1 (odd ladder, $d \geq 15$). *(Proven.) For odd $d \geq 15$, every admissible input satisfies $K^{(d)}(x) \in T_d$.*

Proof. Let Z_0, Z_1, Z_{2+} count the pairs with $\delta_k = 0,\ 1,\ \geq 2$. The blocks contribute $2Z_0 + Z_1$ zero digits, and from $2Z_{2+} + Z_1 \leq \sum_k \delta_k \leq 9$ (Lemma 4.5),

$$2Z_0 + Z_1 \;=\; 2M - Z_1 - 2Z_{2+} \;\geq\; 2M - 9.$$

For odd $d \geq 17$, $M = (d-5)/2 \geq 6$ and the block region alone has ≥ 3 zeros. At $d = 15$, $M = 5$ gives only ≥ 1, and the second zero comes from a case split on x_4:

Case $x_4 = 0$. Sorting forces $x_5 = \cdots = x_{14} = 0$, every $\delta_k = 0$, and the block region is ten zeros.

Case $1 \leq x_4 \leq 7$. The five δ_k are alternate adjacent differences among positions $5, \ldots, 14$, so their sum telescopes below $x_5 - x_{14} \leq x_5 \leq x_4 \leq 7$, whence $2Z_0 + Z_1 \geq 10 - 7 = 3$.

Case $x_4 \in \{8, 9\}$. Sorting forces $x_0, \ldots, x_3 \in \{8, 9\}$, and a direct check of the two subcases $x_0 = x_3$ and $x_0 = x_3 + 1$ gives $\operatorname{core}(x) \leq 999$ and $\operatorname{core}(x) \leq 9999$ respectively — in both, $\operatorname{core}(x) < 10^4$, so decimal place 4 of the output is zero, joining the ≥ 1 block zero. ■

E.2 (even ladder, $d \geq 18$). *(Proven.) For even $d \geq 18$, every admissible input satisfies $K^{(d)}(x) \in T_d$.*

Proof. The even core core_6 occupies places 0–5 (its exact maximum over the 5,005 sorted 6-tuples is $899{,}991 < 10^6$, attained at $(9,9,9,9,9,0)$), so it never carries past place 5, and the appended pairs stack directly above it at places $(6,7), (8,9), \ldots$ (Appendix F). With $M' = (d-6)/2$ pairs the block count gives $2Z_0 + Z_1 \geq 2M' - 9$ exactly as in E.1, which is ≥ 2 as soon as $M' \geq 6$, that is $d \geq 18$. ■

E.3 (the enumerated range). *(Computed; every row re-run in full for this paper.) For $7 \leq d \leq 16$, the maximum number of steps for an admissible orbit to enter T_d, over all admissible multisets on the appropriate ladder, is*

d	admissible multisets	max steps to T_d
7	11,340	8
8	24,210	6
9	48,520	3
10	92,278	3
11	167,860	2
12	293,830	2
13	497,320	2
14	817,090	2
15	1,307,404	1
16	2,042,875	1

*The $d = 15, 16$ rows confirm one-step closure where E.1 applies and just below the even threshold; exhaustive one-step confirmations at $d = 17$ (*3,124,450 *multisets), $d = 18$ (*4,686,725*), and $d = 20$ (*10,014,905*) are in the project log [S].*

E.1–E.3 together give a uniform reaching-time bound at every $d \geq 7$, which is Lemma 4.5's role in the induction. One gap remains: when an orbit lands in T_d and Lemma 4.3 projects it to length $d - 2$, the projected multiset may not be admissible there.

E.4 (projection lemma). *For $n \in A_d \cap T_d$ with sorted form $(x_0, \ldots, x_{d-3}, 0, 0)$, write m for the length-$(d-2)$ projection. Exactly three things can happen. If m is admissible, the inductive hypothesis applies. If m is a repdigit $(v, \ldots, v)$, then n is block-aligned, $K^{(d)}(n) = 0$, and $n \in E_d$. If m is a near-repdigit, the orbit recovers an admissible projection within two further steps and the induction proceeds.*

Proof of the recovery claim. The admissible Case-2 inputs at d have sorted form $(v^{d-3}, w, 0, 0)$ with $v \neq 0$, $w \neq v$ — eighty-one (v, w) pairs — and project to the near-repdigit $\{v^{d-3}, w\}$ at length $d-2$. On a near-repdigit every gap below the top rank vanishes, so each appended pair contributes nothing and the image equals the core, in closed form on either ladder.*

If $w < v$, the top digit is v and $K^{(d-2)} = 9(v-w) \cdot 10^{d-4}$. For $v - w \geq 2$ this is two distinct nonzero digits above $d-4$ zeros — admissible at $d-2$, so the orbit recovers in one step. For $v - w = 1$ it is $9 \cdot 10^{d-4}$, padding to the near-repdigit $(9, 0, \ldots, 0)$; one further step gives $K^{(d-2)} = 9 \cdot 9900 = 89{,}100$, padding to $(9, 8, 1, 0, \ldots, 0)$ — admissible. Recovery in at most two steps.

If $w > v$, the top digit is w and the same cancellation gives $K^{(d-2)} = 9900\,(w - v)$ on both ladders — for the even root, $9+90-9000+99000-99999 = -9900$ collects the v-coefficients to the identical value. This runs through $9900, 19800, \ldots, 79200$ as $w - v = 1, \ldots, 8$, each admissible at $d - 2$ (top digits $9, 9$ when $w - v = 1$, otherwise three distinct nonzero digits) above $d - 4$ zeros. Recovery in one step.

*Only the ladder roots $d - 2 \in \{5, 6\}$, lacking the appended-pair structure, fall outside the closed forms; there the projection reaches 60714 by the census. The closed forms hold at every length, so the induction is closed uniformly in d; direct enumeration confirms them — seventy-two pairs recovering in one step, the nine with $v - w = 1$ in two — together with total reaching times at most 27 steps on the odd ladder and 15 on the even, for every $7 \leq d \leq 401$. ■

F The ladder, explicitly

Labelling sorted positions $a = s_0, b = s_1, c = s_2, \ldots$ in descending order, a rule is two letter-strings whose k-th letter names the position feeding decimal place 10^{d-1-k} of the respective rearrangement. The ladder of Theorem 4.1 reads:

d	ladder	π	σ
5	odd (root)	`adcbe`	`deacb`
6	even (root)	`eadcbf`	`fdeacb`
7	odd	`fgadcbe`	`gfdeacb`
8	even	`hgeadcbf`	`ghfdeacb`
9	odd	`hifgadcbe`	`ihgfdeacb`
10	even	`jihgeadcbf`	`ijghfdeacb`
11	odd	`jkhifgadcbe`	`kjihgfdeacb`
12	even	`lkjihgeadcbf`	`klijghfdeacb`
13	odd	`lmjkhifgadcbe`	`mlkjihgfdeacb`
14	even	`nmlkjihgeadcbf`	`mnklijghfdeacb`
15	odd	`nolmjkhifgadcbe`	`onmlkjihgfdeacb`
16	even	`ponmlkjihgeadcbf`	`opmnklijghfdeacb`
17	odd	`pqnolmjkhifgadcbe`	`qponmlkjihgfdeacb`
18	even	`rqponmlkjihgeadcbf`	`qropmnklijghfdeacb`
19	odd	`rspqnolmjkhifgadcbe`	`srqponmlkjihgfdeacb`
20	even	`tsrqponmlkjihgeadcbf`	`stqropmnklijghfdeacb`

Each rung extends the rung two above by one prepended letter-pair — the appended zero-sum coefficients $(\pm 9 \cdot 10^{d-2}, \mp 9 \cdot 10^{d-2})$, the two ladders differing only in the pair's sign convention — and the four coefficients $(9900, 9, 90, -9000)$ at 60714's nonzero digits recur verbatim in every row. Under the fixed-point assignment $a = 7$, $b = 6$, $c = 4$, $d = 1$, all later letters zero, every row evaluates to $\pi \cdot s = 71460$ and $\sigma \cdot s = 10746$, so $K = |71460 - 10746| = 60714$; each row of the table was machine-checked (Computed).

The recipe is mechanical at any length. At $d = 100$ (even ladder): positions 0–3 carry the locked coefficients $(9900, 9, 90, -9000)$, positions 4–5 the split pair $(99000, -99999)$, and positions 6–99 carry forty-seven appended zero-sum pairs $(\mp 9 \cdot 10^{2k+6}, \pm 9 \cdot 10^{2k+6})$. Every coefficient from position 4 on multiplies a zero digit of the padded fixed point, and

$$K(60714) \;=\; |\,9900 \cdot 7 + 9 \cdot 6 + 90 \cdot 4 - 9000 \cdot 1\,| \;=\; |69300 + 54 + 360 - 9000| \;=\; 60714,$$

the same five-term arithmetic at every length (machine-checked at $d = 100$). What is engineered here is only the fixed-point equation; the basin, which is the theorem, is Section 4 and Appendix E.

References

[Kaprekar 1955] D. R. Kaprekar, "An interesting property of the number 6174," *Scripta Mathematica* **15** (1955), 244–245.

[Trigg 1974] C. W. Trigg, "All three-digit integers lead to 495," *The Mathematics Teacher* **67** (1974), 41–45.

[Prichett 1981] G. D. Prichett, A. L. Ludington, and J. F. Lapenta, "The determination of all decadic Kaprekar constants," *The Fibonacci Quarterly* **19** (1981), 45–52. Precursor: G. D. Prichett, "Kaprekar's routine with five-digit integers," *The Fibonacci Quarterly* **17** (1979), 154–159.

[Peterson 2008] A. Peterson and J. Pulapaka, "Kaprekar's routine and other number sequences in different bases," *Virginia Mathematics Teacher* **35**(1) (2008).

[Nuez 2021] F. Nuez, "Parametric transformation functions in the Kaprekar routine I," arXiv:2110.15123 (2021).

[Nuez 2021b] F. Nuez, "Algebraic Kaprekar routine architecture II," arXiv:2111.00932 (2021).

[Nuez 2022] F. Nuez, "Symmetries in generalized Kaprekar's routine," *Symmetry* **14**(1), 37 (2022).

[Devlin 2021] M. Devlin and T. Zeng, "Maximum distances in the four-digit Kaprekar process," *Integers* **21**, A52 (2021).

[Kay 2022] A. Kay and K. Downes-Ward, "Fixed points and cycles of the Kaprekar transformation: 1. Odd bases," *Journal of Integer Sequences* **25**, Article 22.6.7 (2022).

[Kay 2024] A. Kay and K. Downes-Ward, "Fixed points and cycles of the Kaprekar transformation: 2. Even bases," arXiv:2408.12257 (2024).

[Dahl 2026] C. D. Dahl, "Coarse-grained drift fields and attractor-basin entropy in Kaprekar's routine," *Entropy* **28**, 92 (2026).